Homework 7- Solutions

Problem 1. Let $X_1, X_2, ..., X_{81}$ be a random sample of size $n = 81$ from a distribution with p.d.f.

$$f(x) = \begin{cases} \dfrac{(6-x)^2}{72} & 0 < x < 6 \\ 0 & \text{otherwise} \end{cases}$$

Find $P(\bar{X} < 1.6)$ approximately.

$$\mu = E[X] = \int_{-\infty}^{\infty} x f_X(x)dx = \int_0^6 \frac{x(6-x)^2}{72}dx = \int_0^6 (\frac{1}{2}x - \frac{1}{6}x^2 + \frac{1}{72}x^3)dx = (\frac{1}{4}x^2 - \frac{1}{18}x^3 + \frac{1}{288}x^4)\Big|_0^6 = \frac{3}{2}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)dx = \int_0^6 \frac{x^2(6-x)^2}{72}dx = \int_0^6 (\frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{72}x^4)dx = (\frac{1}{6}x^3 - \frac{1}{24}x^4 + \frac{1}{360}x^5)\Big|_0^6 = \frac{36}{10}.$$

$$\sigma^2 = E[X^2] - (E[X])^2 = \frac{36}{10} - \frac{9}{4} = \frac{72-45}{20} = \frac{27}{20}$$

$$E[\bar{X}] = E[X] = \frac{3}{2} = 1.5$$

$$Var(\bar{X}) = \frac{Var(X)}{n} = \frac{1}{60} \text{ and, } \sigma = \sqrt{Var(\bar{X})} = \frac{1}{\sqrt{60}}$$

$$P(\bar{X} < 1.6) = P(\frac{\bar{X} - E[\bar{X}]}{\sqrt{Var(\bar{X})}} < \frac{1.6 - 1.5}{\frac{1}{\sqrt{60}}}) = P(\frac{\bar{X} - E[\bar{X}]}{\sqrt{Var(\bar{X})}} < (0.1)\sqrt{60}) = \Phi((0.1)\sqrt{60}) \quad (1)$$

Problem 2. Let $X_1, \cdots, X_n$ be a be a random sample from

$$f_\theta(x) = \begin{cases} (\theta+1)x^\theta & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the maximum likelihood estimator (MLE) for θ.

The likelihood function for sample $X_1, \cdots, X_n$ from the above distribution is

$$L(\theta) = \prod_{i=1}^n f_\theta(X_i) = \prod_{i=1}^n (\theta+1)X_i^\theta = (\theta+1)^n \prod_{i=1}^n X_i^\theta$$

Now we take ln of the above function.

$$G(\theta) = ln(L(\theta)) = nln(\theta+1) + \theta \sum_{i=1}^n ln(X_i)$$

Now take derivative of log-likelihood function with respect to θ

$$G'(\theta) = \frac{n}{\theta+1} + \sum_{i=1}^{n} ln(X_i)$$

Now by setting $G'(\theta) = 0$ and solving for θ, we have

$$\hat{\theta}_{MLE} = -1 - \frac{n}{\sum_{i=1}^{n} ln(X_i)}$$

Problem 3. Use the Central limit theorem to approximate the following probabilities. Don't forget to apply the continuity correction (only if necessary).

(a) Suppose we roll a fair 4-sided die until we get 100 threes. What is the probability it takes at least 1000 rolls until this happens?

(b) Let X be the sum of 7500 real numbers, and Y be the same sum, but with each number rounded to the nearest integer before summing. If the fractions rounded off are independent and each one is uniformly distributed over (-0.5, 0.5), use the Central Limit Theorem to estimate the probability that $P(|X - Y| > 50)$.

We know that $X \sim NBin(r = 100, p = \frac{1}{4})$ so $E[X] = \frac{r}{p} = 400$ and $var(X) = \frac{r(1-p)}{p^2} = 1200$. By the CLT $X \approx N(\mu = 400, \sigma^2 = 1200)$. We have to apply the continuity correction since we are approximating a discrete distribution (NBin) with a continuous one (Normal).

$$P(X \geq 1000) = P(X \geq 999.5) = 1 - P(X < 999.5) = 1 - P(\frac{X-400}{\sqrt{1200}} < \frac{999.5-400}{\sqrt{1200}} = \frac{599.5}{\sqrt{1200}}) = 1 - \Phi(\frac{599.5}{\sqrt{1200}}).$$

b. Let $\epsilon_1, \cdots, \epsilon_n \in (-0.5, +0.5)$ be the roundoff errors (n = 7500). Then,

$$E[\epsilon_i] = \frac{-0.1 + 0.1}{2} = 0$$
$$var[\epsilon_i] = \frac{(0.5 - (-0.5))^2}{12} = \frac{1}{12}$$

Let $X - Y = \epsilon = \sum_{i=1}^{7500} \epsilon_i$, $E[\epsilon] = 0$ and $var(\epsilon) = \frac{7500}{12} = 625$. By the CLT, $\epsilon \approx N(\mu = 0, \sigma^2 = 625)$. Note we don't use a continuity correction here since we are approximating a continuous variable.

$$P(|\epsilon| > 50) = 1 - P(|\epsilon| \leq 50) = 1 - P(\frac{|\epsilon - 0|}{25} < \frac{50-0}{25})$$

Let $Z = \dfrac{\epsilon - 0}{25}$, then $Z \approx N(0,1)$ and using 68-95-99.7 rule we have

$$P(|\epsilon| > 50) \approx 1 - P(|Z| \le 2) = 0.05$$

Problem 4. Suppose we have a population which has a geometric distribution with parameter p Take samples $X_1, \cdots, X_n$ from this distribution.

(a) Find the maximum likelihood estimation for p.

(b) Find the method of moments estimation for p.

Given a random sample $X_1, X_2, \ldots, X_n$ from the geometric distribution, the likelihood function is:

$$L(p) = \prod_{i=1}^{n} P(X_i) = \prod_{i=1}^{n} (1-p)^{X_i - 1} p = p^n (1-p)^{\sum_{i=1}^{n}(X_i - 1)}$$

Taking the log-likelihood:

$$\ln L(p) = n \ln p + \left(\sum_{i=1}^{n} (X_i - 1) \right) \ln(1-p)$$

Differentiating with respect to p and setting it to zero:

$$\frac{\partial \ln L(p)}{\partial p} = \frac{n}{p} - \frac{\sum_{i=1}^{n}(X_i - 1)}{1-p} = 0$$

Simplify:

$$\frac{n}{p} = \frac{\sum_{i=1}^{n}(X_i - 1)}{1-p}$$

Solve for p:

$$p = \frac{n}{\sum_{i=1}^{n} X_i}$$

Thus, the MLE for p is:

$$\hat{p}_{\text{MLE}} = \frac{1}{\bar{X}_n}$$

where $\bar{X}_n$ is the sample mean. (The second derivative test will not be graded.)

(b) Method of Moments (MOM)

The expected value of X for the geometric distribution is:

$$E(X) = \frac{1}{p}$$

Equating the sample mean $\bar{X}_n$ to the population mean $E(X)$:

$$\bar{X}_n = \frac{1}{p}$$

Solving for p, we find:

$$\hat{p}_{\text{MOM}} = \frac{1}{\bar{X}_n}$$